

SOHAN SANJEEVA PUTHRAN

AI Engineer · Applied ML Engineer · Data Scientist

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years building production-grade AI systems and scalable ML pipelines. Specialised in LLM applications (LangChain, RAG, HuggingFace, Ollama), MLOps automation, and deploying models that drove 22% accuracy improvements and 30% revenue growth across healthcare and PropTech. Expert in end-to-end ML lifecycle from research to production deployment on AWS/Azure. IBM RAG & Agentic AI certified (2026).

TECHNICAL SKILLS

AI/ML Engineering: LLMs (Ollama, HuggingFace, Claude, GPT-4), LangChain, LangGraph, RAG, FAISS, Transformers, Vector Databases, Agentic AI, CrewAI, AutoGen

Programming: Python, SQL, R, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

MLOps & Deployment: CI/CD, AWS SageMaker, Azure ML, Docker, Airflow, FastAPI, Streamlit, Model Monitoring

ML Techniques: Time-Series Forecasting (LSTM, ARIMA), Anomaly Detection, NLP, Computer Vision, Recommendation Systems

Data Engineering: Snowflake, AWS Lambda, Azure Data Factory, ETL/ELT Pipelines, PySpark, Data Warehousing

Cloud & DevOps: AWS (SageMaker, S3, Lambda), Azure, Git, Agile, Cross-functional Collaboration

Visualisation: Tableau, Power BI, Streamlit, Folium, Interactive Dashboards

WORK EXPERIENCE

Data Scientist | Lundbeck

Jan 2022 – May 2024

Illinois, USA · Global Pharma · \$3B+ Revenue

- Architected end-to-end ML pipeline for time-series forecasting using TensorFlow and Keras, improving forecast accuracy by 22% and driving profitability optimisation across product lines.
- Led MLOps implementation with AWS SageMaker and CI/CD automation, reducing model refresh time by 36% and enabling continuous deployment of production models.
- Engineered automated data pipelines processing 10M+ records daily using Airflow, Python, SQL, and Snowflake, reducing data latency from 3 hours to 15 minutes.
- Designed deep learning anomaly detection system using classification and neural networks, improving issue detection accuracy by 30% and reducing false positives by 45%.

Junior Data Scientist | Airriva

Jun 2021 – Dec 2021

Ohio, USA · PropTech Startup

- Designed and implemented algorithms leveraging occupancy data analysis to detect demand fluctuations, enabling proactive identification of surge events and contributing to increased profitability.
- Developed proprietary algorithm using feature importance analysis to identify key amenities within each region, resulting in a 26% increase in profits.
- Deployed geospatial maps using Folium to analyse occupancy patterns across 4,500+ listings, facilitating comprehensive understanding of regional occupancy dynamics.

Data Scientist Intern | Airriva

Oct 2020 – May 2021

Ohio, USA · PropTech Startup

- Designed and executed a price prediction model for 4,500+ short-term rental listings, achieving an RMSE of 0.17.
- Architected and deployed an Azure ML pipeline to identify critical features impacting price and occupancy, enhancing data-driven decision-making.
- Designed a surge detector using quadratic regression to predict optimal pricing surges, resulting in a 30% increase in profits through timely price adjustments.

Junior Software Engineer | Plexus365

Sep 2017 – Jun 2018

Bengaluru, India · B2B Commodity Trading

- Developed and maintained cross-platform web and mobile applications using Angular 5, .NET Core, and Ionic 3.
- Integrated Google Maps API for geospatial visualisation and real-time location insights, strengthening full-stack product development foundations.

PROJECTS

AI Trading Copilot | LangChain · HuggingFace · yfinance · Streamlit

Live: [ai-trading-copilot.streamlit.app](#) · [GitHub](#) · Mar – May 2026

- Architected a multi-agent AI trading assistant using LangChain, HuggingFace, and GPT-4 to deliver real-time stock analysis, technical indicators, and conversational financial insights through an interactive Streamlit application.
- Built modular agent pipelines leveraging yfinance APIs, custom financial calculators, and LLM-powered reasoning systems to automate market intelligence generation and improve contextual investment insights.

NL2SQL Assistant | *LangChain · HuggingFace (SQLCoder-7B-2) · RAG (BM25) · Streamlit*
Live: [nl-to-sql-assistant.streamlit.app](#) · [GitHub](#) · Jan – Feb 2026

- Built a schema-aware Natural Language to SQL assistant using Python, LangChain, HuggingFace SQLCoder-7B-2, and SQLite to convert plain-English questions into executable SQL queries.
- Developed a RAG-based write mode using BM25 retrieval, human-in-the-loop approval, database backups, and transactional rollback for safer INSERT/UPDATE/DELETE workflows.

AI-Powered Resume Analyzer | *LangChain · FAISS · Sentence Transformers · Streamlit*
[GitHub](#)

- Built intelligent document parsing system using PyPDF2 and regex-based extraction to structure unstructured resume data into analysable components (contact info, skills, experience).
- Developed semantic matching engine leveraging FAISS vector search and sentence-transformers to compute cosine similarity scores between resumes and job descriptions with 85%+ relevance accuracy.

FREELANCE & VOLUNTEER

Lead ML Engineer | **Omdena** *Jun 2024 – Jan 2025*
Remote

- Led computer vision models for rooftop solar panel and obstacle detection using CNNs and object detection frameworks (YOLO, Faster R-CNN), supporting global sustainability initiatives across 10+ countries.
- Built fraud detection models for financial intelligence using ensemble methods and anomaly detection; won 1st place in UN 'Unite Ideas' challenge for anti-corruption technology.
- Managed cross-functional team of 20+ ML engineers across 4 time zones, implementing agile workflows and code review processes.

EDUCATION

Master of Data Science and Artificial Intelligence *Jun 2024 – May 2026*
Campbellsville University, Louisville, KY, USA

Master of Data Science *Aug 2019 – Jun 2021*
Illinois Institute of Technology, Chicago, IL, USA

Bachelor of Computer Science and Engineering *Aug 2013 – Jun 2017*
Visvesvaraya Technological University, Belgaum, India

CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate · IBM / Coursera · January 2026
RAG, LangChain, LangGraph, CrewAI, AutoGen, vector databases, multimodal GenAI, AI agents